PRAVEEN KUMAR

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About

Principal AI Engineer and Solution Architect with 6+ years building and deploying production AI systems across LLMs, speech, document intelligence and enterprise platforms.

Profile

Hands-on engineering lead with ownership of architecture, model evaluation, inference and delivery. Build and deploy Python, PyTorch and FastAPI systems, including multilingual speech, RAG and document workflows, across cloud and on-premise environments.

Skills

Core Expertise

- AI system architecture and delivery across speech analytics, LLM applications, RAG, document processing and multi-tenant platforms.
- Model training, evaluation and deployment, with attention to latency, memory use, data privacy and operational reliability.

Programming & Frameworks

- Languages: Python, SQL, Bash, JavaScript and React; desktop applications with Tauri.
- Frameworks: FastAPI, Flask, Django REST, PyTorch, Hugging Face Transformers, LangGraph, LangChain, OpenAI SDK and Unsloth.
- Inference: vLLM, llama.cpp, Ollama, ONNX Runtime and Triton Inference Server.
- Speech: FasterWhisper, WhisperX, sherpa-onnx, pyannote.audio, speaker diarization and text-to-speech.

Data & Model Engineering

- LoRA/QLoRA fine-tuning, dataset curation, label audits, model evaluation, quantisation and CPU/GPU optimisation.
- Retrieval and vector databases: Qdrant, FAISS, Chroma and PostgreSQL vector search.
- OCR and layout-aware document parsing: PDFPlumber, Surya OCR and PaddleOCR.

Infrastructure & DevOps

- Docker, Docker Compose, NVIDIA Container Toolkit, GPU partitioning and on-premise deployment.
- CI/CD with GitHub Actions and Jenkins; environment management with UV, Conda and Poetry.
- Background processing with Redis Streams and queues, Celery, Nginx and load balancing.
- Cloud deployment: AWS EC2, S3 and ECR; Azure; on-premise CPU/GPU infrastructure.

System Architecture & Backend

- Modular APIs, asynchronous processing, agent workflows, PostgreSQL, Redis and tenant isolation.
- Access controls, JWT, 2FA, secrets management, audit trails and signed desktop updates.

Monitoring, Logging & Observability

- Fluent Bit, Loki, OpenSearch, Grafana and Prometheus for service monitoring and performance analysis.
- Model and service health checks, latency measurement, drift monitoring and release validation.

Leadership & Delivery

- Architecture reviews, engineering mentorship, technical planning and cross-functional delivery.
- Work with product and client teams on requirements, deployment constraints and release readiness.

Experience

TrellisSoft, Inc. | *Bengaluru, India*

Principal AI Engineer | *Jul 2025 - Present*

- Own architecture and core engineering for speech, LLM and document platforms, from requirements and model evaluation through deployment.
- Lead delivery of desktop agent-assist and database intelligence systems, including AI pipelines, integrations and tenant isolation.
- Choose CPU/GPU serving approaches against quality, latency and memory constraints; benchmark alternatives before release.
- Lead architecture reviews, mentor engineers and coordinate deployment readiness with product and client teams.

Senior Artificial Intelligence Engineer | *Jun 2023 - Jul 2025*

- Delivered multilingual transcription, diarization and call-quality automation for contact-centre workflows.
- Built RAG knowledge systems and layout-aware OCR pipelines for insurance and finance, with validation and review workflows.
- Fine-tuned domain models with LoRA/QLoRA and served them with vLLM; implemented retraining and drift monitoring.
- Mentored engineers and standardised AI design reviews; recognised for mentorship and product delivery.

Artificial Intelligence Engineer | *Aug 2020 - Jun 2023*

- Built NLP, speech-recognition and computer-vision systems for chatbots, sentiment analysis and document recognition.
- Implemented data pipelines, feature engineering and automated training, taking models through deployment and monitoring.
- Received Brainiac of the Quarter (2021) and Employee of the Quarter recognition (2022/2023).

Projects

Docuvera360: Database and Document Intelligence
Role: AI Architect and Core Engineer | Trellissoft | May 2026 - Present
Tech: Python, FastAPI, React, PostgreSQL, Redis, DuckDB, Chroma, vLLM and Docker

- Built the successor to Loss360, enabling users to query ERP databases, spreadsheets and documents on their own infrastructure.
- Developed text-to-SQL and hybrid RAG agents, with checks for duplicate joins, mixed currencies and unintended query filters.
- Implemented saved query plans for dashboards and alerts without repeat LLM calls. Measured alert checks fell from 29 s to 0.25 s.
- Supported seven databases, Salesforce, SharePoint and 26 file types, with PostgreSQL tenant isolation and audit controls.
- Benchmarked vLLM at 590 tokens/s across 10 concurrent streams on one 48 GB GPU.

PulseAI360: Live AI Agent Assist
Role: Principal AI Architect and Engineering Lead | Trellissoft | Apr 2026 - Present
Tech: Python, FastAPI, Tauri, React, sherpa-onnx, ONNX Runtime, llama.cpp and Redis

- Designed the platform and wrote most of its code with a small team. The ongoing early-beta product supports live contact-centre agents with guidance, category tracking and post-call work.
- Built desktop audio capture, multilingual transcription and local CPU inference for agent workstations, with deployment and signed-update support.
- Integrated call guidance, summaries, AI evaluation, sentiment, critical flags and compliance checks into the agent workflow.
- In laptop UAT, transcribed a 6.4-second speech turn in 406 ms. Added 1,394 tests to support release validation.

Kavach PII 270M: Multilingual PII Detection and Redaction
Role: Independent Model Developer | Sep 2026 - Sep 2026 | Completed
Tech: Python, PyTorch, Transformers, PEFT/LoRA and llama.cpp

- Fine-tuned and published a 270M-parameter Gemma 3 model covering 43 personal-data entity types across 29 languages.
- Built data generation and label audits, LoRA training, evaluation, source-text matching and timestamp-preserving redaction.
- Version 3 achieved 92.0 value-F1 on a 4,577-document held-out set and 90.4 on a separate 480-document code-generated probe, with published test conditions and limitations.
- Released a 196 MB GGUF build for local inference. Model and evaluation details.

Education

MVJ College of Engineering
Aug 2016 - Jul 2020

- Computer Science and Engineering.
- GPA: 8.0/10.

Achievements

- GitHub sponsorship from Recall.ai for open-source contributions.
- Excellence Appreciation, July 2025, for product contributions and leadership.
- Mentorship and Coaching Award, January 2024.
- Excellent Project Deliverables Award, 2023, for Employee Training and OCR systems.

Habit/Interest

- Technical writing and editing for Analytics Vidhya on Medium; football and FIFA.

Portfolio | GitHub | LinkedIn